

Peng: Streaming a 311B Model on Consumer Hardware

A Colibri-style MoE engine running Xiaomi's MiMo-V2.5 entirely via NVMe streams.



The Legend of the Peng and the Hardware Reality

The Chinese Zhuangzi myth describes a colossal bird that stays airborne exclusively by riding a continuous whirlwind.

Peng-MIMO applies this exact physical constraint to consumer AI execution: an impossibly large model kept aloft by an unrelenting stream of disk reads.

47 MoE layers × 8 active experts
× 12.6 MB = ~4.7 GB
cold reads required per single token

The Peng Metaphor Diagram



Colibri Architecture Splits Residency by Access Frequency

Mixture of Experts (MoE) models activate only a fraction of their parameters per token. In MiMo-V2.5, only ~15B parameters do the work.

The engine maps this mathematical sparsity directly to physical hardware residency.

The Colibri Residency Map

Resident in RAM (Int4)
Dense Layer 0 (MLP 16384),
Attention blocks,
Embeddings, Learned
Expert PPIN cache



Resident on NVMe (~165 GB) —
12,032 routed experts
(47 layers × 256 experts)

**Active Compute
Membrane** —
Per-Layer LRU Cache
moving 12.6 MB expert
blocks on demand

Isolating the Perfect Vessel for Desktop Streaming

Maximizing **Colibri reuse** required a model with massive **overall scale** but **minimal active parameters**, compatible router math, and manageable attention mechanics.

Model Selection Matrix

	MiMo-V2.5	GLM-5.2 (744B)	Hunyuan Hy3 (295B)	Qwen3.5 (122B)
Disk Int4 Size	~165 GB (Fits 256GB SSD)	~370 GB ✗ (Fails constraints)	~157 GB	~65 GB
Active Params per Token	15B Active	40B Active	21B Active	10B Active
Router Compatibility	Identical to Colibri	Identical	Different ⚠	Different ⚠
Attention Simplicity	Simple GQA	MLA+DSA	GQA, Heterogeneous	Gated DeltaNet (C-hostile)
Checkpoint Format	FP8 128×128	-	Different	Different

THE LOGICAL GEOMETRY OF MiMo-V2.5

A 48-layer structure heavily optimized for sparse activation. **QKV** is fused in a single tensor, and **V** is scaled $\times 0.707$ before attention. The vocabulary utilizes a 152,576 byte-BPE style matrix.

THE SPINE



THE ACTIVE COMPUTE BUDGET



HYBRID ATTENTION SINKS CONTEXT COSTS

A severe risk to 32GB RAM systems is KV-cache bloat during long context generation. MiMo-V2.5 prevents this via an aggressive 9-to-39 hybrid attention ratio.

The Sinking Attention Diagram



Takeaway: Long context comes almost free in RAM compared to full-attention architectures.

THE WHIRLWIND TOKEN GENERATION FLOW

Every expert ID lookup triggers a cascading search through physical memory layers. Prefetching and PILOT routines only change when bytes move, not the underlying math.

VRAM (GPU-Pin): Upload dense tensors + ~522 hottest experts. Runs moe_acc for decode.

Miss

RAM (Learned Pin): Host keeps .coli_usage learned cache. Pins most-used experts into remaining DDR4/DDR5.

Miss

RAM (Per-Layer LRU): Transient cache catching recent expert drops.

Miss

Disk (NVMe Pread): Cold Miss. Multi-GB read triggered. Fetches 12.6 MB Int4 block per expert.

BIT-EXACT TOKENS: PROVING THE MATH BEFORE THE SPEED

Before streaming a 152GB container, the C-engine (`c/mimo.c`) must **perfectly reproduce** the **reference transformers** implementation, accounting for Range-TP interleaved QKV layouts and **partial RoPE dimensions**.

Tiny Oracle

Generated with
official
`modeling_mimo_v2.py`
(real hybrid pattern,
sink bias)

C Engine Matching

Must hit
Teacher-Forcing
32/32 and Greedy
20/20 against the
Oracle without a
single bit flip.

Lossless Packing

Int8 dequant vs
packed Int4
verification.

Full Real Container

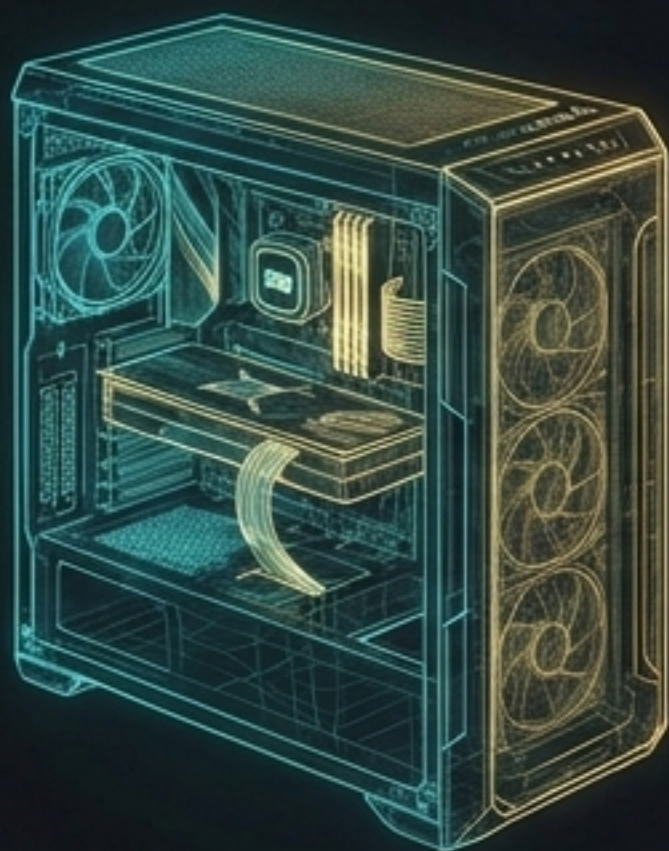
316 GB FP8
converted to 152
GB Int4. Booting
the real model.

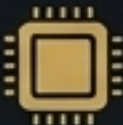
THE PHYSICS OF THE COLD START

Operating on a standard consumer desktop, the sheer volume of NVMe reads dictates the baseline speed. Without warm caches, the disk is the absolute ceiling.

Spec & Benchmark Dashboard

THE RIG



Intel Xeon W-2140B
(8C/16T, AVX2) 

23 GB Available RAM 

RTX 3060 12GB 

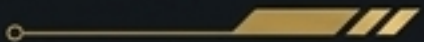
WSL2 ext4 (/root) 

THE PHYSICS

MAX DISK SPEED: 2.75 GB/s (O_DIRECT measured)

COST PER TOKEN: ~4.7 GB reads 

COLD BASELINE: 0.31 TOK/S

HIT-RATE: 44% 

With only 22GB RAM, the engine fits ~3 experts per layer (cap drops 64→3).
Every token requires re-reading experts. The system is entirely disk-bound.

BREAKING THE SPEED CEILING

At extreme RAM starvation, overlapping reads and predicting router behavior is the only way forward.

PILOT Prefetching



PILOT=1 pre-fetches experts based on router lookahead. **Speedup: 1.3x to 2.0x** in RAM-starved setups.

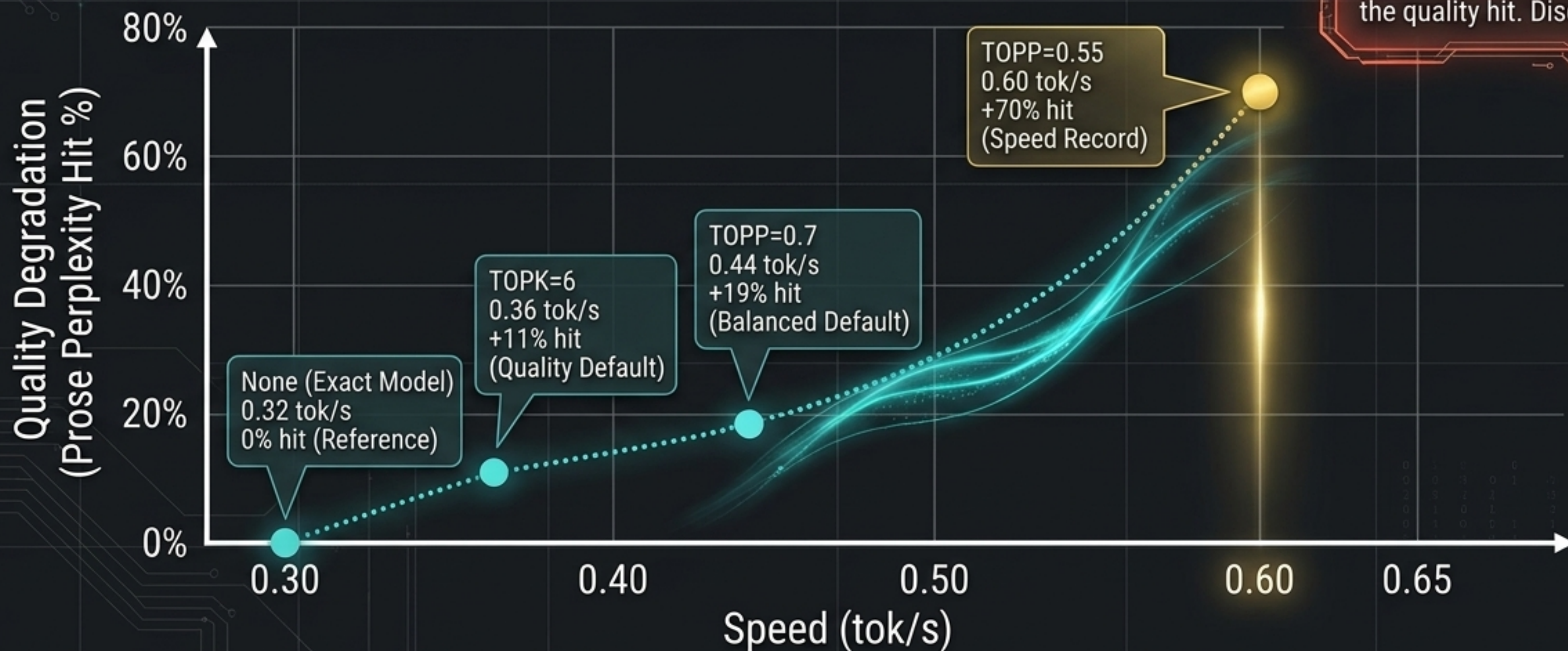
The 0.60 Record

0.60 tok/s

- Conditions: Autopin + GPU-First Pin + TAO=1 / SPEED=1
- Hit-rate: ~54%
- Profile: Disk ~14s · Attn ~9s · Matmul ~4.6s

THE QUALITY VS. SPEED FRONTIER

The 0.60 tok/s record requires skipping low-router-weight experts. Trimming forms a strict frontier—speed tracks the number of experts loaded almost linearly, but perplexity degrades.



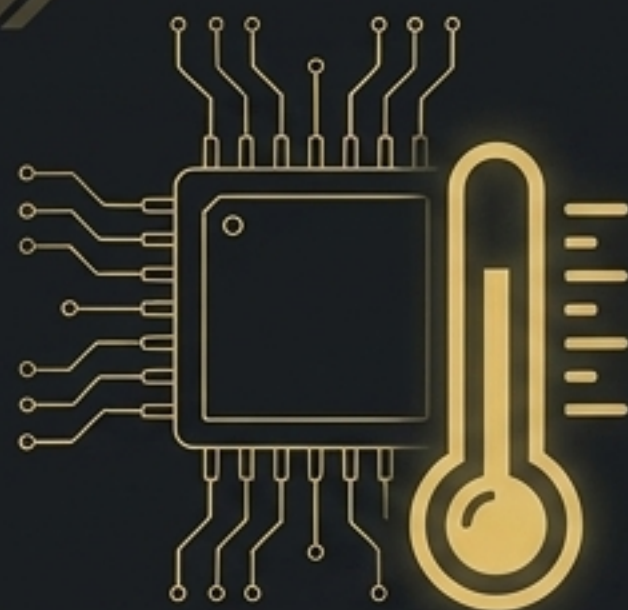
THE REALITY OF NVME ENDURANCE

Streaming a 152GB model continuously raises immediate concerns about SSD death. However, read streams interact with flash memory physics differently than page writes.



Terabytes Written (TBW) vs Reads

Consumer NVMe endurance is rated in TBW. Multi-TB read streams do not consume TBW. Writes from .coli_usage logs are physically negligible.



Thermal Throttling

Sustained generation will heat the drive controller. Heat, not write-wear, is the primary physical constraint for long benchmarks.



OS Intercepts

Do not stream over /mnt/c VHDX boundaries on Windows. The 9p protocol destroys I/O. Must stream from local ext4/NVMe.

CORRIENTE PENG: THE RESIDUAL STREAM AS A RIVER

The engine's startup scripts (TA0=1) eschew brute force for proportional tuning. The system treats the network not as a static math equation, but as fluid dynamics carving channels through rock.



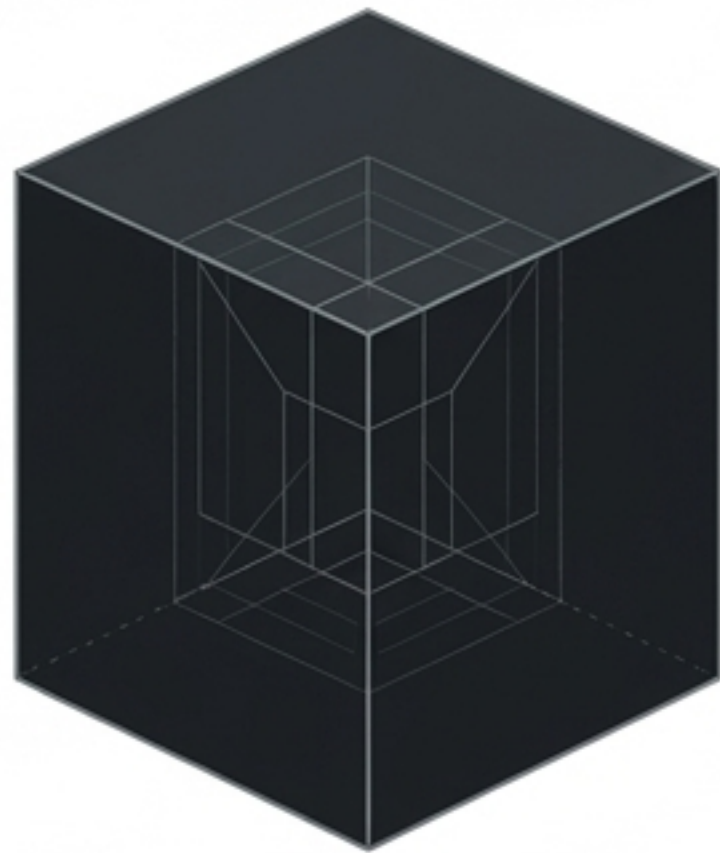
SACRED PROPORTIONS

- **ENERGY:** Uses ϕ (Golden Ratio) of free VRAM to allocate buffers.
- **FLOW:** Fibonacci-based parameter tuning.
- **HABIT:** .coli_traj tracks network routing, digging memory channels for future expert pins.

THE DESKTOP FRONTIER: NAVIGATION OVER LOADING

Peng-MIMO proves that frontier-class intelligence (311B parameters) does not require massive datacenter VRAM. It only requires a shift in engineering philosophy.

THE OLD PARADIGM: THE MONOLITH



- Treats models as monolithic objects.
- Must load 100% of weights into VRAM to begin generation.
- Hits an absolute hard-stop at memory limits.

THE PENG PARADIGM: THE LANDSCAPE

- Treats models as vast geographic landscapes.
- Keeps only the core (dense) resident. NVMe reads navigate the MoE terrain on demand.
- Trades absolute speed for infinite physical capability.



TELEGRAM PATIENCE, FRONTIER QUALITY. GATE 1.0 TOK/S IS STILL OPEN.